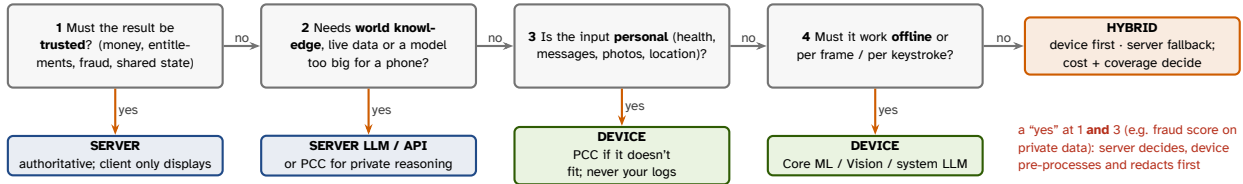


**In one line:** Place each computation where its **binding constraint** is cheapest: **trust** (anything authoritative runs on a server — the client is attacker-controlled), **knowledge and size** (world knowledge, big models, live data → cloud), **privacy, offline and latency** (personal data, per-frame work, no network → device). Most real features are **hybrid**: device first, server (or Private Cloud Compute) as the fallback, results cached and **versioned**.

#### Ask in this order — the first “yes” decides



| factor                  | on-device                                                | Private Cloud Compute (Apple)                           | your server / cloud LLM                                   |
|-------------------------|----------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
| latency                 | no network; first model load is slow — load early, reuse | network round trip                                      | round trip + queue; radio wake-up on cellular             |
| cost per call           | free to you; paid in the user's battery + heat           | Apple's servers; quota + entitlement (iOS 27 API)       | you pay per request / GPU hour, scales with users         |
| privacy                 | data never leaves the phone                              | leaves, but stateless, no privileged access, verifiable | leaves: consent, retention, region, subprocessors         |
| offline model / quality | ✓<br>small (system LLM ≈3B, 2-bit; your Core ML model)   | ✗<br>much larger; 32K-token context, reasoning          | ✗<br>any size, frontier models, tools, RAG over your data |
| freshness               | updates with the OS / app / asset pack                   | Apple-managed                                           | instant; live data; world knowledge                       |
| coverage                | AI-capable devices only; memory + thermal limits         | Apple Intelligence devices only (unverified)            | every device, Android, web — one behaviour                |
| logic security          | untrusted: code + weights extractable, results forgeable | not your server — no custom logic                       | authoritative; secrets stay here                          |
| observability           | no server logs; on-device evals, opt-in telemetry        | none for you                                            | full logs, A/B, evals — mind PII                          |

#### The options

- **Apple on-device:** Foundation Models (system LLM, offline, free; check availability); Core ML / Core AI for *your* model; Vision, Natural Language, Speech for common tasks — try these first.
- **Android on-device:** Gemini Nano runs in the **AICore** system service; apps use **ML Kit GenAI** APIs (Prompt, Summarization, Proofreading, Rewriting, Image Description, Speech Recognition). AICore ships and updates the model — no model in your APK.
- **Private Cloud Compute:** Apple's server extension of on-device AI. Five requirements — **stateless** computation, **enforceable** guarantees, **no privileged runtime access**, **non-targetability**, **verifiable transparency**. The device encrypts only to nodes whose **attested** software is in a public **transparency log**; an **OHTTP relay** run by a third party hides the IP.
- **Your server:** any model, tools and data; you own cost, keys, logs and legal basis. API keys never ship in the app — the app calls *your* backend, which calls the model.

#### Hybrid patterns

- **Device first, server fallback:** unavailable device, context too long, low confidence, guardrail refusal → escalate (with consent).
- **Router / cascade:** a cheap on-device classifier decides which requests are “hard” and need the big model.
- **Split:** device pre-processes — OCR, speech-to-text, embeddings, **PII redaction**, downscaling — and sends less, safer data.
- **Server trains, device infers:** ship or download the model; personalise on device (Core ML MLUpdateTask); only the model moves, not the data.
- **Precompute + cache:** shared results (recommendations, translations of shared content) computed once on the server, cached on devices.
- **Speculative:** show the on-device draft now, replace it if the server's answer arrives and differs — label which one the user sees.

#### Traps

- ! “On-device is free” — battery, heat, memory (jetsam), app size, device coverage.
- ! Validating a purchase, score or discount on the client — anything on the device can be patched; the server decides.
- ! A vendor LLM key in the app “to save a backend” — extracted in minutes; proxy it.
- ! Server fallback that silently uploads personal data — fallback needs consent + redaction.
- ! No unavailable path: the system model is missing on older or ineligible devices.

#### Example — device first, server fallback

```
protocol Summarizer: Sendable {
    func summarize(_ text: String) async throws -> String
}
struct HybridSummarizer: Summarizer {
    let local: any Summarizer // Foundation Models session
    let remote: any Summarizer // YOUR backend, not a vendor key
    let consented: @Sendable () -> Bool
    func summarize(_ text: String) async throws -> String {
        if SystemLanguageModel.default.isAvailable,
            text.count < 6_000 { // rough fit; 26.4+ tokenCount(for:)
            do { return try await local.summarize(text) }
            catch { /* context / guardrail / refusal: escalate */ }
        }
        guard consented() else { throw SummaryError.needsConsent }
        return try await remote.summarize(redactPII(text))
    }
}
```

#### Model delivery and results

- **Bundle** the model (app size, every user pays) or **download** it: Background Assets managed asset packs (essential before launch, prefetch, or on demand; optional Apple hosting), then `MLModel.compileModel(at:)` for a Core ML file.
- Store **model id + version** beside every output; cache results keyed by `hash(input) + modelVersion`; a new model invalidates, never silently mixes.
- Results are *data*: sync them through the normal sync path (private → the user's devices only). Compute **shared** results once on the server, **private** ones per device.
- Same feature, different engines → different answers on iOS, Android, old phones: design the UI for variable quality, evaluate each path separately.

#### Remember

**Trust** → **server** · **knowledge** → **cloud** · **private, offline, per-frame** → **device** · **everything else: device first, cloud fallback, version the results.**